Curve Sketching（曲線描繪）



你能描繪 (sketch) 以下函數之圖像嗎？

(a) $y = x^3 - 6x^2 + 9x - 1$

(b) $y = \dfrac{1}{x+2}$

(c) $y = \dfrac{x^2+1}{x^2-1}$

(d) $y = \dfrac{x^3}{(x-1)^2}$

✓ 若「是」的話，你便不用學此課了！

✘ 若「否」的話，你便要學怎樣用微分去描繪 $y = f(x)$ 的圖像了。

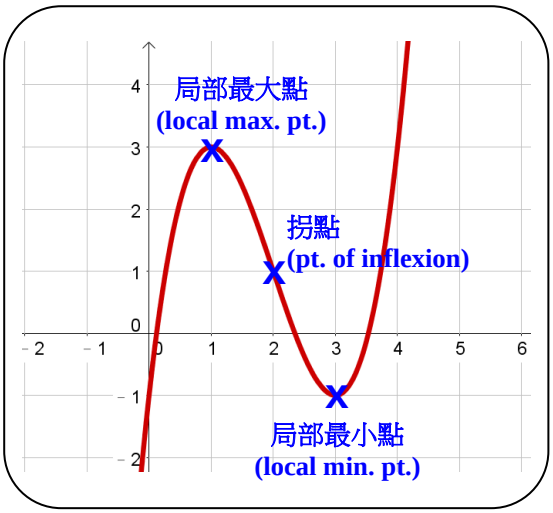


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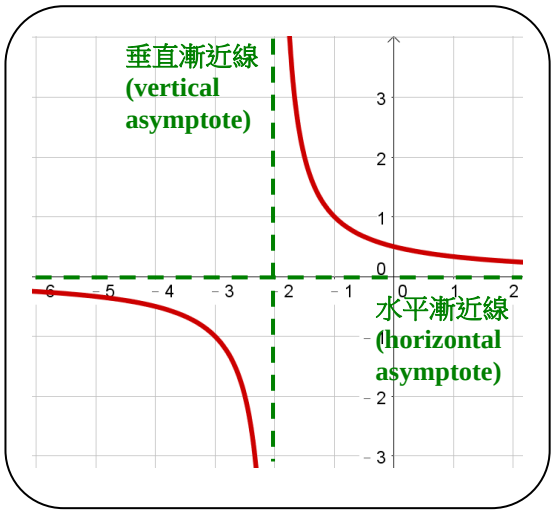
Curve Sketching (曲線描繪)

曲線描繪 (Curve sketching) 其實除了要用曲線將函數之圖像繪畫出來外，還要標示圖像的一些特徵 (如有) 才算是完成，以下是一些例子：

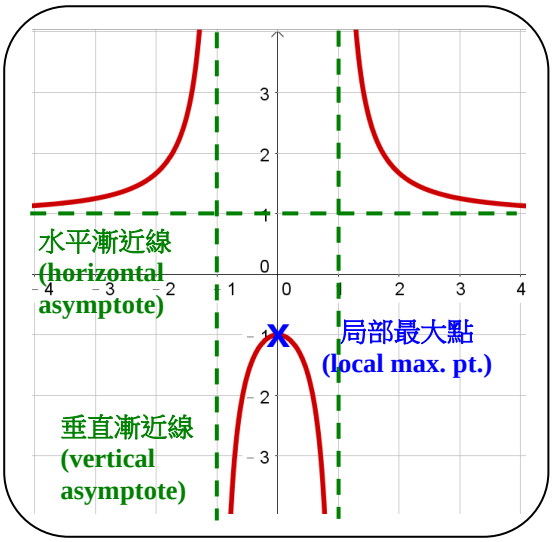
(a) $y = x^3 - 6x^2 + 9x - 1$



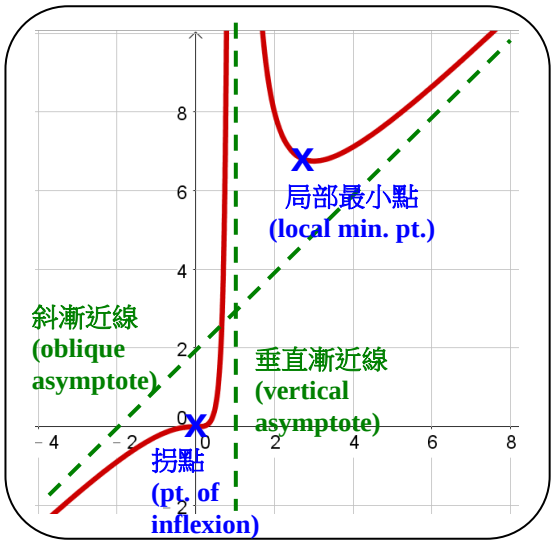
(b) $y = \dfrac{1}{x+2}$



(c) $y = \dfrac{x^2+1}{x^2-1}$



(d) $y = \dfrac{x^3}{(x-1)^2}$

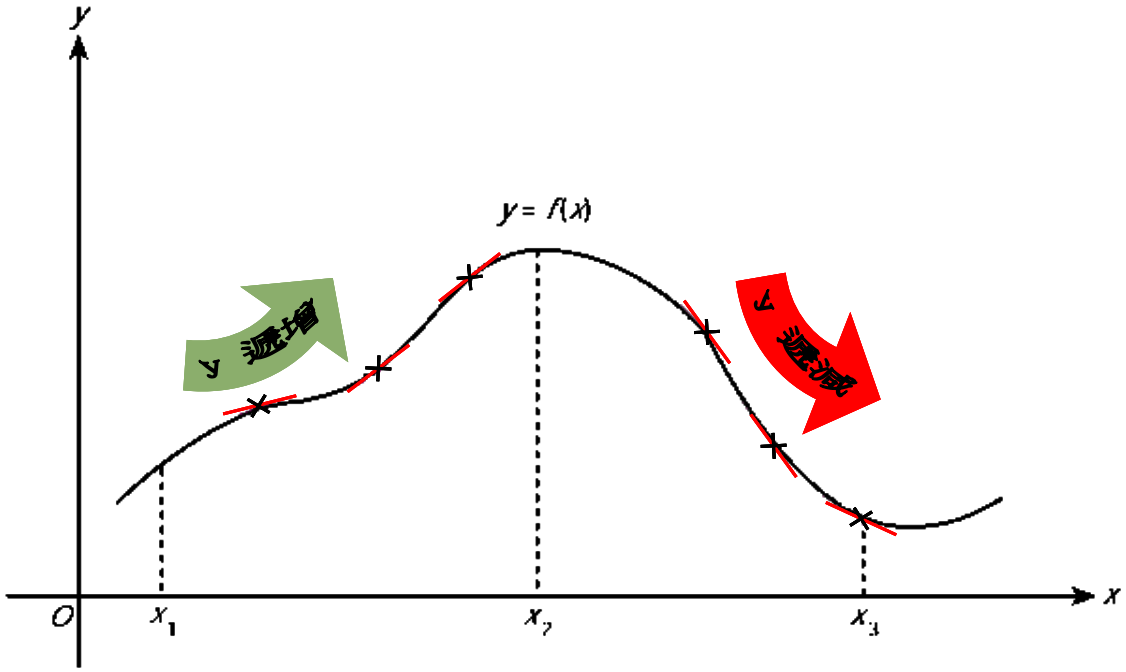


接下來請按部就班地學習如何完成曲線描繪吧！

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Increasing / Decreasing Functions (遞增/遞減函數)

以下圖像顯示 $y = f(x)$ 的圖像：



當 $x_1 < x < x_2$

- $y = f(x)$ 的值是遞增 **(increasing)**
- $y = f(x)$ 的斜率 **(Slope)** 是正值
- $\frac{dy}{dx} > 0$

當 $x_2 < x < x_3$

- $y = f(x)$ 的值是遞減 **(decreasing)**
- $y = f(x)$ 的斜率 **(Slope)** 是負值
- $\frac{dy}{dx} < 0$

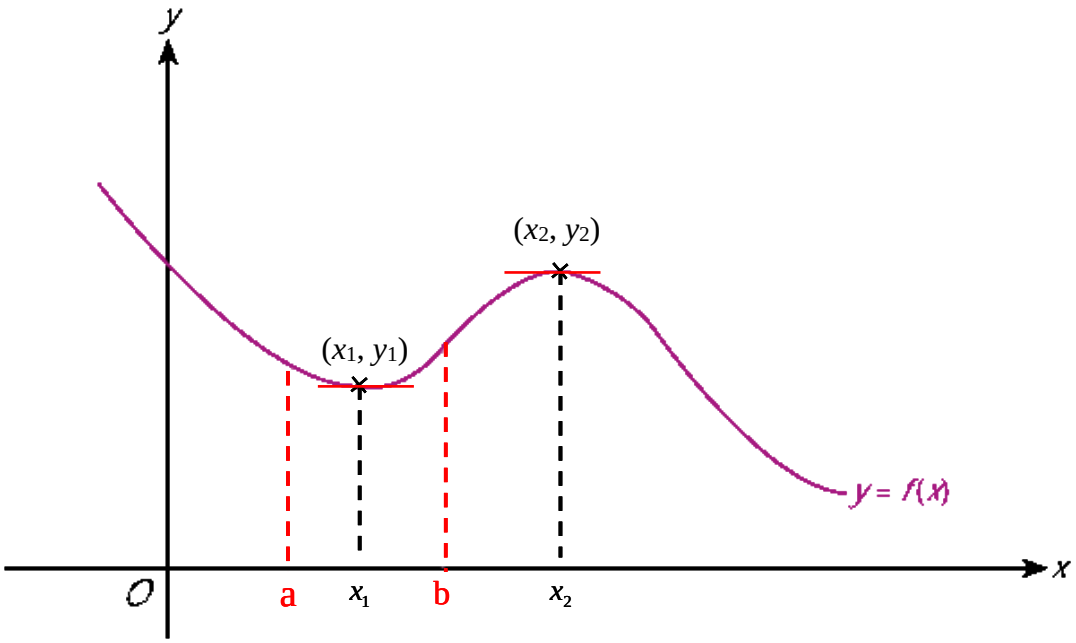
所以，在區間 $a < x < b$ 中：

- 若 $\frac{dy}{dx} > 0$, 則 $y = f(x)$ 是遞增 ***(increasing)***.
- 若 $\frac{dy}{dx} < 0$, 則 $y = f(x)$ 是遞減 ***(decreasing)***.

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Stationary Points (駐點)

當曲線上某點 (x_1, y_1) 的斜率是零的話，該點便被稱為**駐點** **(stationary point)** :



在上述例子中， (x_1, y_1) 及 (x_2, y_2) 是曲線 $y = f(x)$ 的駐點 (stationary points)。

總括而言，在曲線 $y = f(x)$ 上的任何一點 (x_1, y_1) ，

$$\left.\frac{dy}{dx}\right|_{x=x_1} = 0 \quad \Rightarrow \quad (x_1, y_1) \text{ 是駐點 } (\textit{stationary point}).$$

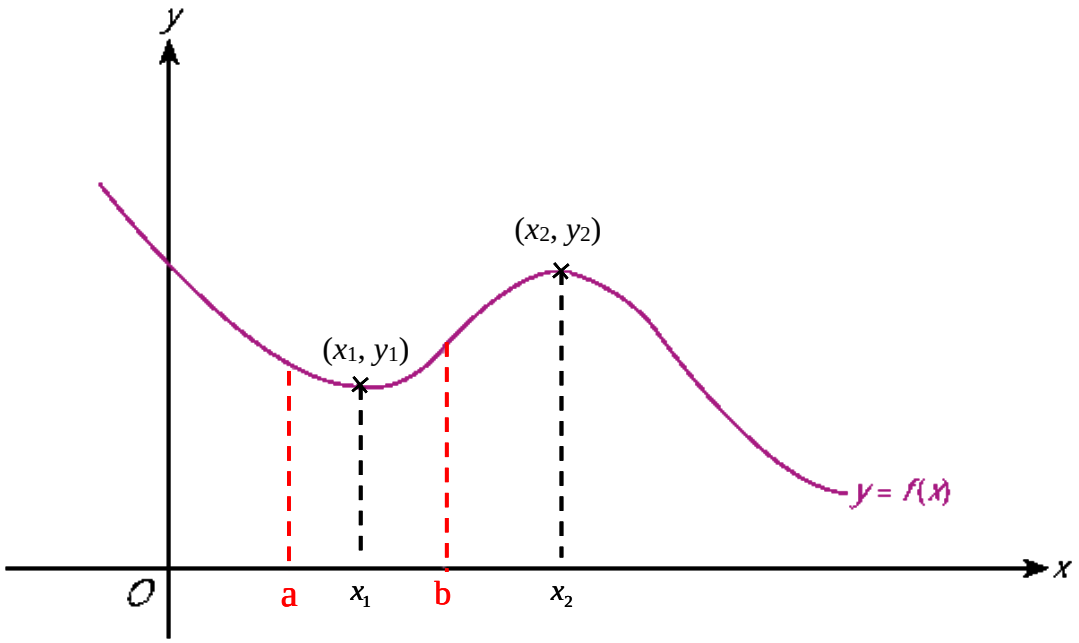
Remark:

曲線上的駐點可以是

- 局部極小值 Local Minimum (試考慮 $y = x^2$)
- 局部極大值 Local Maximum (試考慮 $y = -x^2$)
- 兩者都不是 neither Local Max or Local Min (試考慮 $y = x^3$)

local Maximum / Minimum（局部極大/極小值）

仍是以上頁的曲線 $y = f(x)$ 為例，

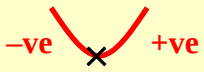


當 $x = x_1$ 時，相比相鄰 (neighbor) 的點 $x = a$ and $x = b$，$y = f(x)$ 的值是最小的。

我們說　　當 $x = x_1$ 時，$y = f(x)$ 有 局部極小值 **(local minimum)**

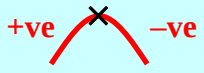
當 $x = x_2$ 時，$y = f(x)$ 有 局部極大值 **(local maximum)**

(x_1, y_1) 是局部極小點 *(local min point)*



若 $\left.\frac{dy}{dx}\right|_{x=x_1} = 0$ 及 $\frac{dy}{dx}$ 的值於經過 $x = x_1$ 時由負變正

(x_1, y_1) 是局部極大點 *(local max point)*



若 $\left.\frac{dy}{dx}\right|_{x=x_1} = 0$ 及 $\frac{dy}{dx}$ 的值於經過 $x = x_1$ 時由正變負

***極值點 (turning point) 及局部極值 (local extrema) 均是局部極大點或局部極小點之綜合稱號。**

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First Derivative Test （一階導數判別法）

e.g. Find the stationary points of the graph $y = 2x^3 - 9x^2 - 60x + 32$.

$$\frac{dy}{dx} = 6x^2 - 18x - 60$$

Set $\frac{dy}{dx} = 0$, i.e. $6x^2 - 18x - 60 = 0$

$x = -2$ or 5

For $x = -2$, $y = 2(-2)^3 - 9(-2)^2 - 60(-2) + 32 = 100$

For $x = 5$, $y = 2(5)^3 - 9(5)^2 - 60(5) + 32 = -243$

$\therefore$ The stationary points are $(-2, 100)$ and $(5, -243)$

Technique

首先設 $\frac{dy}{dx} = 0$

求曲線之所有駐點

(stationary points)

接著我們可以用 **一階導數判別法 (First Derivative Test)** 去判別所求得的駐點是 **局部極大點 (local max. pt.)** 還是**局部極小點 (local min. pt.)**，如下表所示：

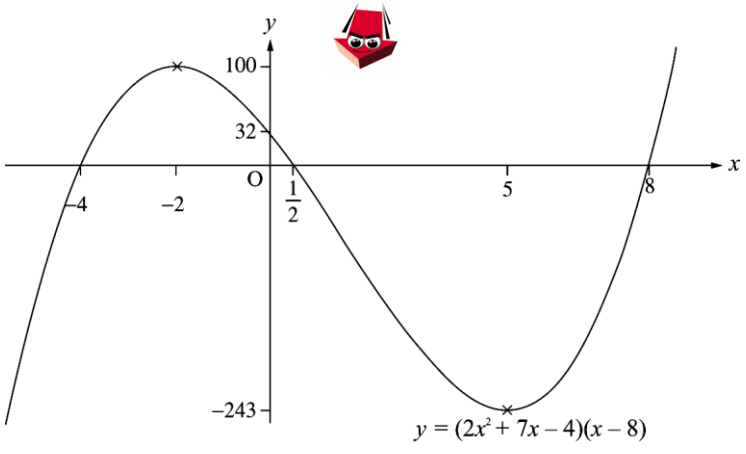
stationary point　　stationary point

	$x < -2$	$x = -2$	$-2 < x < 5$	$x = 5$	$x > 5$
$\frac{dy}{dx}$	+ve	0	–ve	0	+ve

$\frac{dy}{dx}$ **changes from +ve to –ve** through $x = -2$ $\therefore$ $(-2, 100)$ is **local max. pt.**

$\frac{dy}{dx}$ **changes from –ve to +ve** through $x = 5$ $\therefore$ $(5, -243)$ is **local min. pt.**

除了局部極大點與局部極小點外，我們還要求得 x 軸截距及 y 軸截距 (x- and y-intercepts)，才能繪出一個較詳細的曲線圖像。



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First Derivative Test

[A.Math 1995 Paper I Q.3]

Using the information in the following table, sketch the graph of $y = f(x)$, where $f(x)$ is a polynomial.

	$x < 0$	$x = 0$	$0 < x < 1$	$x = 1$	$1 < x < 2$	$x = 2$	$x > 2$
$f(x)$		1		2		1	
$f'(x)$	< 0	0	> 0	0	< 0	0	> 0

(5 marks)

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First Derivative Test



[A.Math 2010 Q.2]

It is given that $f(x)$ is a polynomial with the following properties:

[1] $f(0) = f(2) = f(4) = 0$;

[2] For $0 \le x \le 4$, the minimum and maximum values of $f(x)$ are -2 and 2 respectively;

[3]

	$0 \le x < 1$	$x = 1$	$1 < x < 3$	$x = 3$	$3 < x \le 4$
$f'(x)$	>0	0	<0	0	>0

Using the above information, sketch the graph of $y = f(x)$ for $0 \le x \le 4$.

(4 marks)

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First Derivative Test



[A.Math 2008 Q.13]

Let $f(x) = x(x-6)^2$.

(a) Find the maximum and minimum points of the graph of $y = f(x)$.

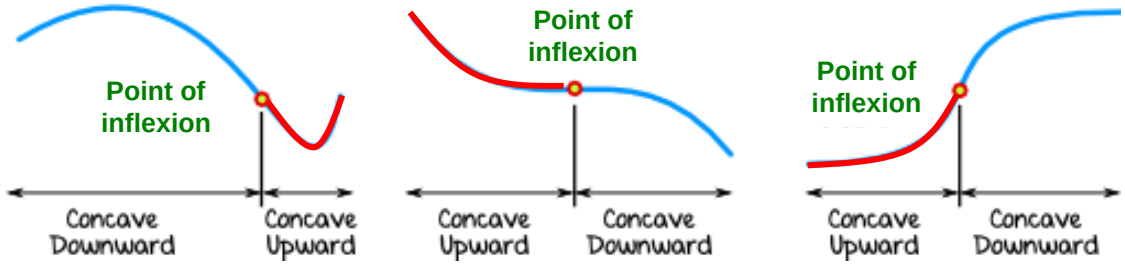
(b) Sketch the graph of $y = f(x)$. ***(7 marks)***

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Second Derivative Test（二階導數判別法）

有時候使用**二階導數判別法（Second Derivative Test）**去判別駐點是局部極大點或是局部極小點會較為方便。在區間 $a < x < b$ 中：

- 若 $\frac{d^2y}{dx^2} > 0$，則 $y = f(x)$ 是開口向上 ***(concave upward).***
- 若 $\frac{d^2y}{dx^2} < 0$，則 $y = f(x)$ 是開口向下 ***(concave downward).***
- 若 $\frac{d^2y}{dx^2}$ 的值於經過 $x = x_1$ 時由正變負 ，則 (x_1, y_1) 是拐點 ***(point of inflexion).***
  (或負變正)



怎樣用二階導數判別法？

例如承上題，我們已得知 (–2, 100) 及 (5, –243) 是駐點 (詳見上頁)，

使用二階導數判別法 (Second Derivative Test),

For $x = -2$, $\frac{d^2y}{dx^2} = 12(-2) - 18 = -42 < 0$ (con. down)

∴ (–2, 100) is a local max. pt.



For $x = 5$, $\frac{d^2y}{dx^2} = 12(5) - 18 = 42 > 0$ (con. up)

∴ (5, –243) is a local min. pt.



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Second Derivative Test



[A.Math 1991 Paper I Q.4]

Let $y = x + \sin 2x$ where $0 \le x \le \pi$

(a) Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.

(b) Find the maximum and minimum values of y. ***(7 marks)***

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Second Derivative Test



[M2 2018 Q.8]

Define $f(x)=\dfrac{A}{x^2-4x+7}$ for all real numbers x, where A is a constant. It is given that the extreme value of $f(x)$ is 4.

(a) Find $f'(x)$.

(b) Find the point(s) of inflexion of the graph of $y = f(x)$. ***(6 marks)***

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Second Derivative Test



[M2 2013 Q.5]

Consider a continuous function $\mathrm{f}(x)=\dfrac{3-3x^2}{3+x^2}$. It is given that

x	$x<-1$	-1	$-1<x<0$	0	$0<x<1$	1	$x>1$
$\mathrm{f}'(x)$	+	+	+	0	−	−	−
$\mathrm{f}''(x)$	+	0	−	−	−	0	+

('+' and '–' denote 'positive value' and 'negative value' respectively.)

(a) Find all the maximum and/or minimum point(s) and point(s) of inflexion.

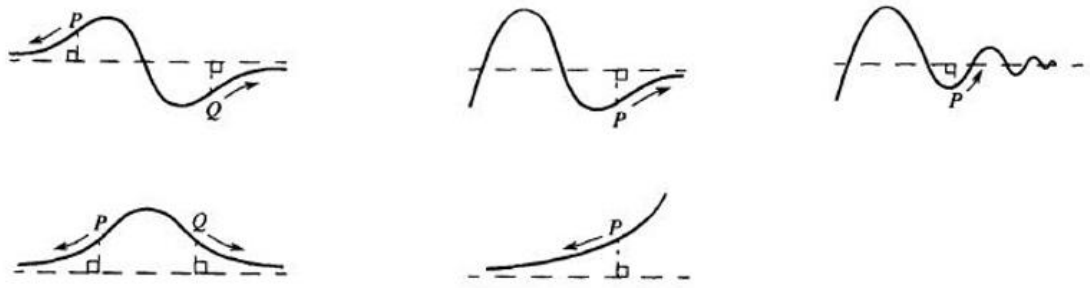
(b) Sketch graph of $y=\mathrm{f}(x)$. ***(6 marks)***

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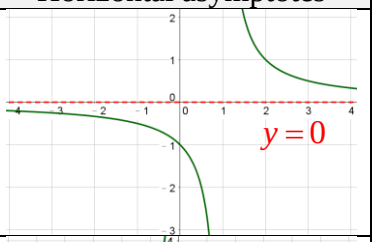
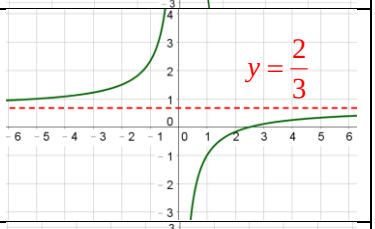
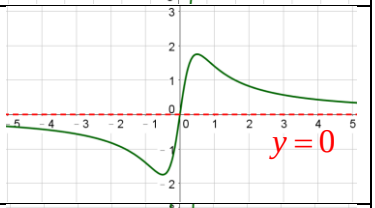
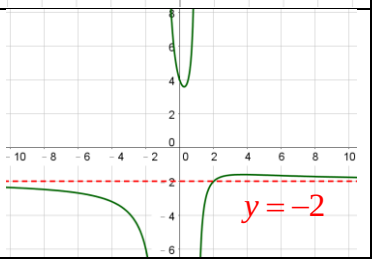
Horizontal Asymptotes (水平漸近線)

若 $\lim_{x\to\infty} f(x) = k$ (或 $\lim_{x\to-\infty} f(x) = k$)，則 $y = k$ 是圖像的水平漸近線 **(horizontal asymptotes).**

水平漸近線的例子如下：



以下是常見有水平漸近線的函數：

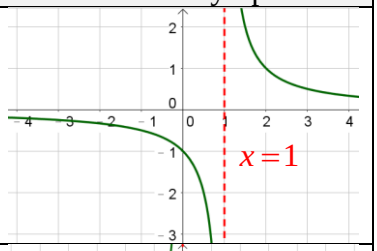
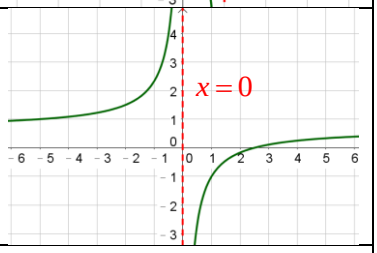
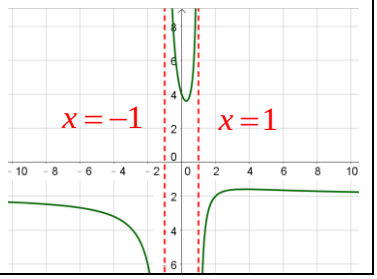
	Function	Limit tends to infinity	Horizontal asymptotes
1.	$f(x) = \dfrac{1}{x-1}$	$\lim\limits_{x\to\infty} f(x) = \lim\limits_{x\to\infty} \dfrac{\frac{1}{x}}{1-\frac{1}{x}} = 0$	
2.	$g(x) = \dfrac{2x-5}{3x}$	$\lim\limits_{x\to\infty} g(x) = \lim\limits_{x\to\infty} \dfrac{2-\frac{5}{x}}{3} = \dfrac{2}{3}$	
3.	$h(x) = \dfrac{7x}{4x^2-1}$	$\lim\limits_{x\to\infty} h(x) = \lim\limits_{x\to\infty} \dfrac{\frac{7}{x}}{4-\frac{1}{x^2}} = 0$	
4.	$f(x) = \dfrac{-2x^2+3x-4}{x^2-1}$	$\lim\limits_{x\to\infty} f(x) = \lim\limits_{x\to\infty} \dfrac{-2+\frac{3}{x}-\frac{4}{x^2}}{1-\frac{1}{x^2}} = -2$	

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Vertical Asymptotes（垂直漸近線）

若 $\lim_{x\to k} f(x) = \pm\infty$ ，則 $y = k$ 是圖像的**垂直漸近線 (vertical asymptotes).**

垂直漸近線多數出現於函數未能定義 (undefined) 的 x 值，以下是一些常見例子：

	Function	Limit tends to k	Horizontal asymptotes
1.	$f(x) = \dfrac{1}{x-1}$ (undefined at $x = 1$)	$\lim_{x\to 1} f(x) = \infty$	
2.	$g(x) = \dfrac{2x-5}{3x}$ (undefined at $x = 0$)	$\lim_{x\to 0} g(x) = \infty$	
3.	$f(x) = \dfrac{-2x^2+3x-4}{x^2-1}$ $= \dfrac{-2x^2+3x-4}{(x+1)(x-1)}$ (undefined at $x = \pm 1$)	$\lim_{x\to 1} f(x) = \infty$ and $\lim_{x\to -1} f(x) = \infty$	

More examples:

Complete the following table:

	Function	Horizontal asymptote(s)	Vertical asymptote(s)
1.	$f(x) = \dfrac{3}{x^2-4}$		
2.	$g(x) = \dfrac{2x}{x^2+1}$		
3.	$h(x) = \dfrac{x^3}{x^3+1}$		
4.	$f(x) = \dfrac{2x-4}{x^2-5x-6}$		

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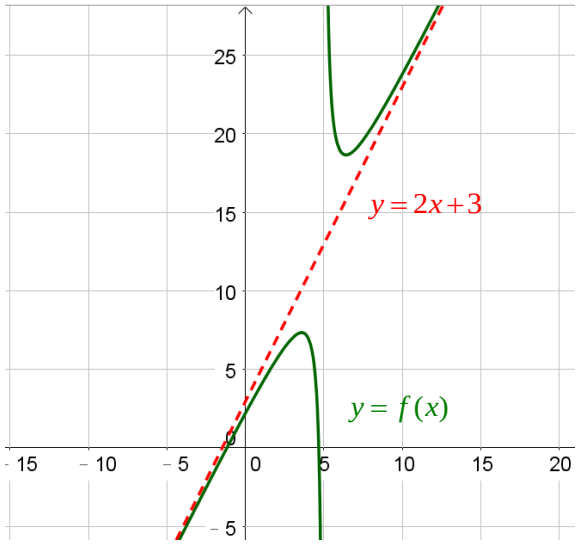
Oblique Asymptotes（斜漸近線）

考慮 $f(x)=2x+3+\dfrac{4}{x-5}$ ：

當 x 接近無限的時候，$\dfrac{4}{x-5}$ 的數值會接近零 (即是 $\lim\limits_{x\to\infty}\dfrac{4}{x-5}=0$)。

那時候 $f(x)$ 的值會接近 $2x+3$ 的值 (即是 $\lim\limits_{x\to\infty}[f(x)-(2x+3)]=0$)。

換言之，$y=2x+3$ 就是函數 $f(x)=2x+3+\dfrac{4}{x-5}$ 圖像的斜漸近線。



以下便是斜漸近線的定義：

若 $\lim\limits_{x\to\infty}[f(x)-(mx+c)]=0$ ，則 $y=mx+c$ 是圖像的**斜漸近線 (oblique asymptotes).**

More examples:

Complete the following table:

	Function	oblique asymptote(s)
1.	$f(x)=3x+4+\dfrac{3}{x^2-4}$	
2.	$g(x)=5x-7-\dfrac{2x}{x^2+1}$	
3.	$h(x)=x^2-3x+4+\dfrac{5}{x+6}$	

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Oblique Asymptotes

若函數是以 $f(x)=\dfrac{P(x)}{Q(x)}$ 表示，其中 $P(x)$ 及 $Q(x)$ 為多項式且 $P(x)$ 的次數比 $Q(x)$ 的次數大 1 (即 $\deg(P(x))-\deg(Q(x))=1$)，則我們需要用長除法去找出斜漸近線。

e.g.1 Find the oblique asymptote of the graph $y = \dfrac{2x^2+3x-4}{x-2}$.

By long division,

$$\frac{2x^2+3x-4}{x-2} = 2x+7+\frac{10}{x-2}$$

$$\begin{array}{r} 2x+7 \\ x-2\,\overline{)\,2x^2+3x-4} \\ \underline{2x^2-4x} \\ 7x-4 \\ \underline{7x-14} \\ 10 \end{array}$$

$\therefore \lim\limits_{x\to\infty}[y-(2x+7)]=0$

$\therefore$ The oblique asymptote is $y = 2x + 7$

Note!

請熟習多項式長除法的做法而不要依賴計算機程式 !!!

e.g.2 Find the oblique asymptote of the graph $y = \dfrac{x^3-3x^2-4}{x^2+1}$.

By long division,

$$\frac{x^3-3x^2-4}{x^2+1} = x-3+\frac{-x-1}{x^2+1}$$

$$\begin{array}{r} x-3 \\ x^2+0x+1\,\overline{)\,x^3-3x^2+0x-4} \\ \underline{x^3+0x^2+x} \\ -3x^2-x-4 \\ \underline{-3x^2+0x-3} \\ -x-1 \end{array}$$

$\therefore \lim\limits_{x\to\infty}[y-(x-3)]=0$

$\therefore$ The oblique asymptote is $y = x - 3$.

Asymptotes



[M2 2012 Q.5]

Find the minimum point(s) and asymptote(s) of the graph of $y = \dfrac{x^2 + x + 1}{x + 1}$.

(6 marks)

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Asymptotes



[M2 2017 Q.9(a)]

Define $f(x)=\dfrac{x^2-5x}{x+4}$ for all $x \neq 4$. Denote the graph of $y = f(x)$ by G.

(a) Find the asymptote(s) of G. ***(3 marks)***

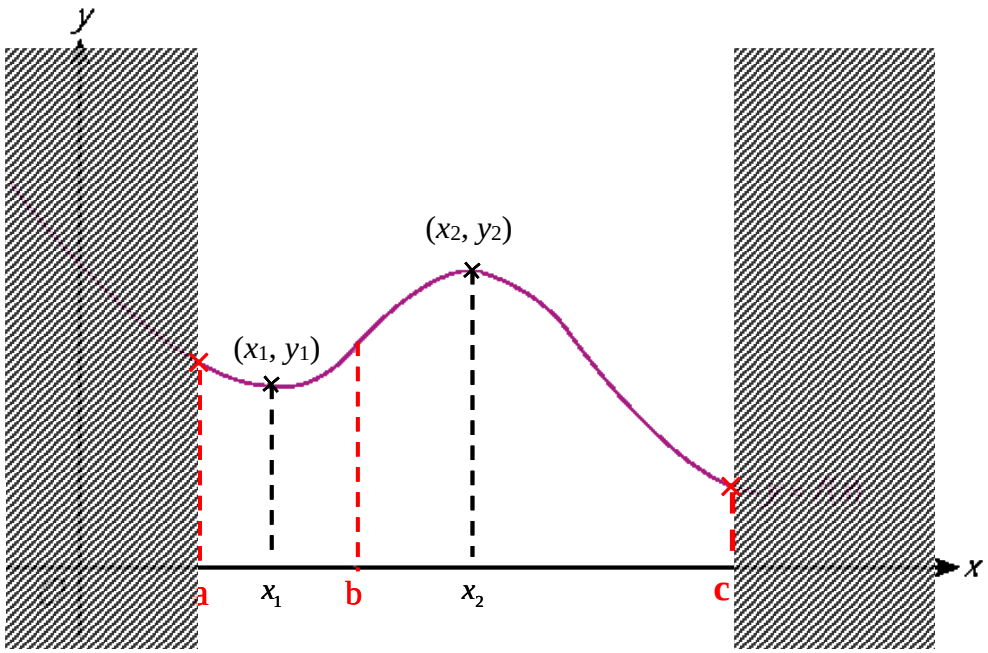
(b) Find $f'(x)$. ***(2 marks)***

(c) Find the maximum and minimum point(s) of G. ***(4 marks)***

Not for sale

Global Maximum / Minimum (全局極大值 / 極小值)

試考慮 $y = f(x)$ 在區間 $a \le x \le c$ 的圖像 (如下圖示) :



在區間 $a \le x \le c$ 中，我們可以見到

- $y = f(x)$ 有 全局極大值（**global maximum**）於 $x = x_2$; 而
- $y = f(x)$ 有 全局極小值（**global minimum**）於 $x = c$.

> **於區間 $a \le x \le c$ 中，全局極大點 / 極小點 (*global maximum / minimum point*) 必定會於以下地點出現 :**
>
> 1. 局部極小點 / 極大點 (*Local Min / Max* pt.) 或
> 2. 終點 *End points* ($x = a$ 或 $x = c$)

* 請保持良好的做數習慣，當已給函數有附帶區間 (interval) 時，請務必去求全局 極大點 / 極小點 (global maximum / global minimum)。



Sketching of Curves（曲線描繪）

6 Steps for curve sketching 描繪曲線六部曲:

Step 1: 求 $y = f(x)$ 的 ***x* 截距 (*x*-intercept(s))** 及 ***y* 截距 (*y*-intercept)**

➔ 求 x 截距： 代 $y = 0$;

➔ 求 y 截距： 代 $x = 0$.

Step 2: 設 $\frac{dy}{dx} = 0$ 去求圖像的**駐點 (stationary point(s))**，並用一階導數判別法 (First Derivative Test) 或 二階導數判別法 (Second Derivative Test) 去求圖像的**局部極大點 (local max pt(s).)** 及**局部極小點 (local min pt(s).)**.

Step 3: 設 $\frac{d^2y}{dx^2} = 0$ 去求圖像的**拐點 (point(s) of inflexion).**

Step 4: 求圖像的**水平 / 垂直 / 斜漸近線 (horizontal / vertical / oblique asymptote(s)).**

Step 5: 求 **全局極大點 (global maximum point)** 及 **全局極小點 (global minimum point.)**.

Step 6: 描繪曲線 (Sketch the curve). 緊記將所有 ***Step 1*** 至 ***Step 5*** 的資料標注在圖中！

Sketching of Curves



[A.Math 1990 Paper I Q.10]

C_1 is the curve $y = \dfrac{2x+1}{x^2+2}$.

(a) (i) Find the range of values of x for which the slope of C_1 is negative

(ii) Find the turning point(s) of C_1, stating whether each point is a maximum or a minimum point. ***(8 marks)***

(b) Sketch the curve C_1 for $-3 \le x \le 3$. ***(4 marks)***

Not for sale



Sketching of Curves

Not for sale

HKCEE Questions (level 1)

1. ***[A.Math 2003 Q.13]***

Let f(x) = 2 sin x – x for $0 \le x \le \pi$. Find the greatest and least value of f(x).

(7 marks)

Not for sale

HKCEE Questions (level 1)

2. ***[A.Math 1988 Paper I Q.4]***

Let $y = \sin x - 2\cos x$ where $0 \le x \le 2\pi$.

(a) Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.

(b) Find the minimum value of y. ***(6 marks)***

Not for sale

HKCEE Questions (level 2)

3. ***[A.Math 2000 Paper I Q.10]***

Let $f(x) = \dfrac{7-4x}{x^2+2}$.

(a) **(i)** Find the x- and y-intercepts of the curve $y = f(x)$.

(ii) Find the range of values of x for which $f(x)$ is decreasing.

(iii) Show that the maximum and minimum values of $f(x)$ are 4 and $-\dfrac{1}{2}$ respectively. ***(9 marks)***

(b) Sketch the curve $y = f(x)$ for $-2 \le x \le 5$. ***(3 marks)***

Not for sale

HKCEE Questions (level 2)

Not for sale

HKCEE Questions (level 2)

4. ***[A.Math 1997 Paper I Q.10]***

The function f(x) = $\dfrac{x^2+kx+9}{x^2+1}$, where k is a constant, attains a stationary value at $x = 3$.

(a) Find f'(x) in terms of k and x

Hence show that $k = -6$. ***(4 marks)***

(b) **(i)** Find the x and y-intercept of the curve y = f(x).

(ii) Find the maximum and minimum points of the curve y = f(x).

(7 marks)

(c) Sketch the graph of y = f(x) for $-6 \le x \le 6$. ***(3 marks)***

Not for sale

HKCEE Questions (level 2)

Not for sale

HKCEE Questions (level 2)

5. ***[A.Math 1998 Paper I Q.10]***

Let $f(x) = 2\cos 2x + 4\sin x - 3$, where $-\pi \le x \le \pi$.

(a) **(i)** Find the x- and y-intercepts of the curve $y = f(x)$.

(ii) Find the maximum and minimum points of the curve $y = f(x)$.

(10 marks)

(b) Sketch the curve $y = f(x)$ for $-\pi \le x \le \pi$. ***(3 marks)***

Not for sale

HKCEE Questions (level 2)

Not for sale

HKDSE Questions (level 3)

6. ***[M2 2015 Q.9]***

Define f(x) = $\dfrac{x^2+12}{x-2}$ for all $x \neq 2$.

(a) Find $f'(x)$. ***(2 marks)***

(b) Prove that the maximum value and the minimum value of f(x) are -4 and 12 respectively, ***(4 marks)***

(c) Find the asymptote(s) of the graph of y = f(x). ***(3 marks)***

HKDSE Questions (level 3)

Not for sale

HKAL Questions (Level 3)

7. ***[P.Math 2002 Paper II Q.8]***

Let $f(x) = x^2 - \dfrac{8}{x-1}$ $(x \neq 1)$.

(a) Find $f'(x)$ and $f''(x)$. ***(2 marks)***

(b) Determine the range of value of *x* for each of the following cases:

(i) $f'(x) > 0$,

(ii) $f'(x) < 0$,

(iii) $f''(x) > 0$

(iv) $f''(x) < 0$ ***(3 marks)***

(c) Find the relative extreme point(s) and point(s) of inflexion of $f(x)$. ***(2 marks)***

(d) Find the asymptote(s) of the graph of $f(x)$. ***(1 mark)***

(e) Sketch the graph of $f(x)$. ***(2 marks)***

Not for sale

HKAL Questions (level 3)

Not for sale

HKAL Questions (level 3)

8. ***[P.Math 2006 Paper II Q.7]***

Let $f(x) = \dfrac{x^2 - x - 6}{x + 6}$ $(x \neq -6)$.

(a) Find $f'(x)$ and $f''(x)$. ***(2 marks)***

(b) Solve each of the following inequalities:

(i) $f'(x) > 0$,

(ii) $f'(x) < 0$,

(iii) $f''(x) > 0$,

(iv) $f''(x) < 0$. ***(3 marks)***

(c) Find the relative extreme point(s) of the graph of $y = f(x)$. ***(2 marks)***

(d) Find the asymptote(s) of the graph of $y = f(x)$. ***(3 marks)***

(e) Sketch the graph of $y = f(x)$. ***(2 marks)***

HKAL Questions (level 3)

Not for sale

HKAL Questions (level 3)

9. ***[P.Math 2000 Paper II Q.9]***

Let $f(x) = \dfrac{x}{(1+x^2)^2}$.

(a) Find $f'(x)$ and $f''(x)$. ***(2 marks)***

(b) Determine the value of x for each of the following cases:

(i) $f'(x) > 0$,

(ii) $f''(x) > 0$. ***(3 marks)***

(c) Find all relative extreme points, points of inflexion and asymptotes of $y = f(x)$. ***(4 marks)***

(d) Sketch the graph of $f(x)$. ***(3 marks)***

Not for sale

HKAL Questions (level 3)

Not for sale

HKAL Questions (Level 4)

10. ***[P.Math 1996 Paper II Q.8]***

Let $f(x) = \dfrac{(x-1)^3}{(x+1)^2}$.

(a) Find $f'(x)$ and $f''(x)$ for $x \neq -1$. ***(2 marks)***

(b) Determine the values of x for each of the following cases:

(i) $f'(x) > 0$,

(ii) $f'(x) < 0$,

(iii) $f''(x) > 0$,

(iv) $f''(x) < 0$. ***(3 marks)***

(c) Find the relative extreme point(s) and point(s) of inflexion of $f(x)$. ***(2 marks)***

(d) Find all asymptotes of the graph of $f(x)$. ***(2 marks)***

(e) Sketch the graph of $f(x)$. ***(2 marks)***

Not for sale

HKAL Questions (level 4)

Not for sale

HKAL Questions (Level 4)

11. ***[P.Math 2007 Paper II Q.7]***

Let f(x) = $\dfrac{(x+15)(x+1)^2}{(x-6)^2}$ $(x \neq 6)$.

(a) Find $f'(x)$ and $f''(x)$. ***(3 marks)***

(b) Solve each of the following inequalities:

(i) $f'(x) > 0$,

(ii) $f''(x) > 0$. ***(2 marks)***

(c) Find the relative extreme point(s) and point(s) of inflexion of the graph y = f(x). ***(4 marks)***

(d) Find the asymptote(s) of the graph of $y = f(x)$. ***(3 marks)***

(e) Sketch the graph of $f(x)$. ***(3 marks)***

Not for sale

HKAL Questions (level 4)

Not for sale

HKAL Questions (level ∞)

*12. ***[P.Math 1997 Paper II Q.8]***

Let $f(x)=\dfrac{x^{\frac{2}{3}}}{x+1}$ $(x\neq -1)$.

(a) **(i)** Find $f'(x)$ for $x\neq -1,0$.

(ii) Show that $f''(x)=\dfrac{2(2x^2-8x-1)}{9x^{\frac{4}{3}}(x+1)^3}$ for $x\neq -1,0$. ***(4 marks)***

(b) Determine the values of x for each of the following cases:

(i) $f'(x)>0$,

(ii) $f'(x)<0$,

(iii) $f''(x)>0$,

(iv) $f''(x)<0$. ***(4 marks)***

(c) Find all relative extrema and points of inflexion of f(x). ***(3 marks)***

(d) Find the asymptote(s) of the graph of f(x). ***(1 mark)***

(e) Sketch the graph of f(x). ***(3 marks)***

Not for sale

HKAL Questions (level ∞)

Not for sale

HKAL Questions (level ∞)

*13. ***[P.Math 2008 Paper II Q.7]***

Let $f(x) = (2x^2 - 14x + 25)e^{2x}$.

(a) Find $f'(x)$ and $f''(x)$. ***(2marks)***

(b) Solve each of the following inequalities:

(i) $f(x) > 0$,

(ii) $f'(x) > 0$,

iii) $f''(x) > 0$. ***(3 marks)***

(c) Find the pont(s) of inflexion of the graph $y = f(x)$. ***(2 marks)***

(d) Find the asymptote(s) of the graph of $y = f(x)$. ***(2 marks)***

(e) Sketch the graph of $f(x)$. ***(2 marks)***

Not for sale

HKAL Questions (level ∞)

Not for sale



More Questions in Past Papers

1. ***[A.Math 1992 Paper I Q.12(a)]***

C_1 is the curve $y = 2\cos 2x - 4\sin x + 1$, where $0 \le x \le \pi$

(i) Find the x- and y-intercepts of C_1.

(ii) Find the turning point(s) of C_1.

(iii) Sketch the curve C_1. ***(12 marks)***

2. ***[A.Math 1996 Paper I Q.9]***

C_1 is the curve $y = \dfrac{4x-3}{x^2+1}$.

(a) (i) Find the x and y - intercepts of the curve C_1;

(ii) Find the range of values of x for which $\dfrac{4x-3}{x^2+1}$ is decreasing;

(iii) Find the turning point(s) of C_1, stating whether each point is a maximum or a minimum point. (Testing for maximum/minimum is not required)

(9 marks)

(b) Sketch the curve C_1 for $-10 \le x \le 10$. ***(3 marks)***

3. ***[A.Math 1986 Paper I Q.10]***

The graph of the function $f(x) = x^3 + hx^2 + kx + 2$ (h and k are constants) has 2 distinct turning points and intersects the line $y = 2$ at the point (0, 2) only.

(a) Show that $3k < h^2 < 4k$. ***(8 marks)***

(b) It is also known that the graph of $f(x)$ passes through (–2, 0).

(i) Express k in terms of h.

(ii) For $h = 4$, find the maximum and minimum points of the graph of $f(x)$ and sketch this graph. ***(8 marks)***

4. ***[M2 2016 Q.9]***

Let a and b be constants. Define $f(x) = x^3 - ax^2 + bx + 5$ for all real numbers x. Denote the curve $y = f(x)$ by C. It is given that $P(-1, 10)$ is a turning point of C.

(a) Find a and b. (3 marks)

(b) Is P a maximum point of C? Explain your answer. (2 marks)

(c) Find the minimum value(s) of $f(x)$. (2 marks)

(d) Find the point(s) of inflexion of C. (2 marks)

5. ***[A.Math 1994 Paper I Q.9]***

Given the curve C: $y = \dfrac{x^2}{1+x} - \dfrac{4}{3}$, where $x \neq 1$.

(a) Find the x and y-intercepts of the curve C. ***(2 marks)***

(b) Find $\dfrac{dy}{dx}$ and show that $\dfrac{d^2y}{dx^2} = \dfrac{2}{(1+x)^3}$. ***(3 marks)***

(c) Find the turning point(s) of the curve C.
For each turning point, test whether it is a maximum or a minimum point.
(6 marks)

(d) Sketch the curve C for $-5 \le x \le 3$. ***(5 marks)***

6. ***[P.Math 2010 Paper II Q.7]***

Let $f(x) = \dfrac{x-1}{(x+3)^3} \quad (x \neq -3)$.

(a) Find $f'(x)$ and $f''(x)$. ***(2 marks)***

(b) Solve
(i) $f(x) > 0$,
(ii) $f'(x) > 0$,
(iii) $f''(x) > 0$. ***(3 marks)***

(c) Find the relative extreme point(s) and point(s) of inflexion of the graph of $y = f(x)$. ***(2 marks)***

(d) Find the asymptote(s) of the graph $y = f(x)$. ***(2 marks)***

(e) Sketch the graph of $f(x)$. ***(3 marks)***

(f) Let $n(k)$ be the number of points of intersection of the graph of $y = f(x)$ and the horizontal line $y = k$. Using the graph of $y = f(x)$, find $n(k)$ for any real values of k. ***(3 marks)***